Report on Practical 4

Router Config using CLI Commands

1. Aims

- Learn to configure networks in in Cisco Packet Tracer using CLI commands to establish connectivity between two PCs through a router.

2. Objectives

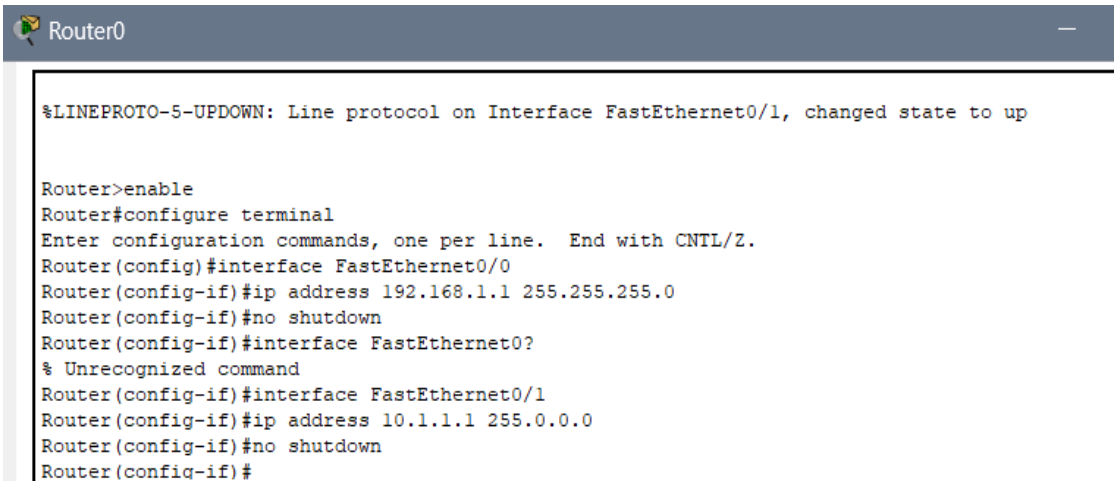
- Understand and apply network configuration concepts learned from the provided video tutorials.
- Configure the router's interfaces with valid IP addresses.
- Configure the host IP addresses for the PCs
- Check connectivity between the PCs using the **ping** command.

3. Background

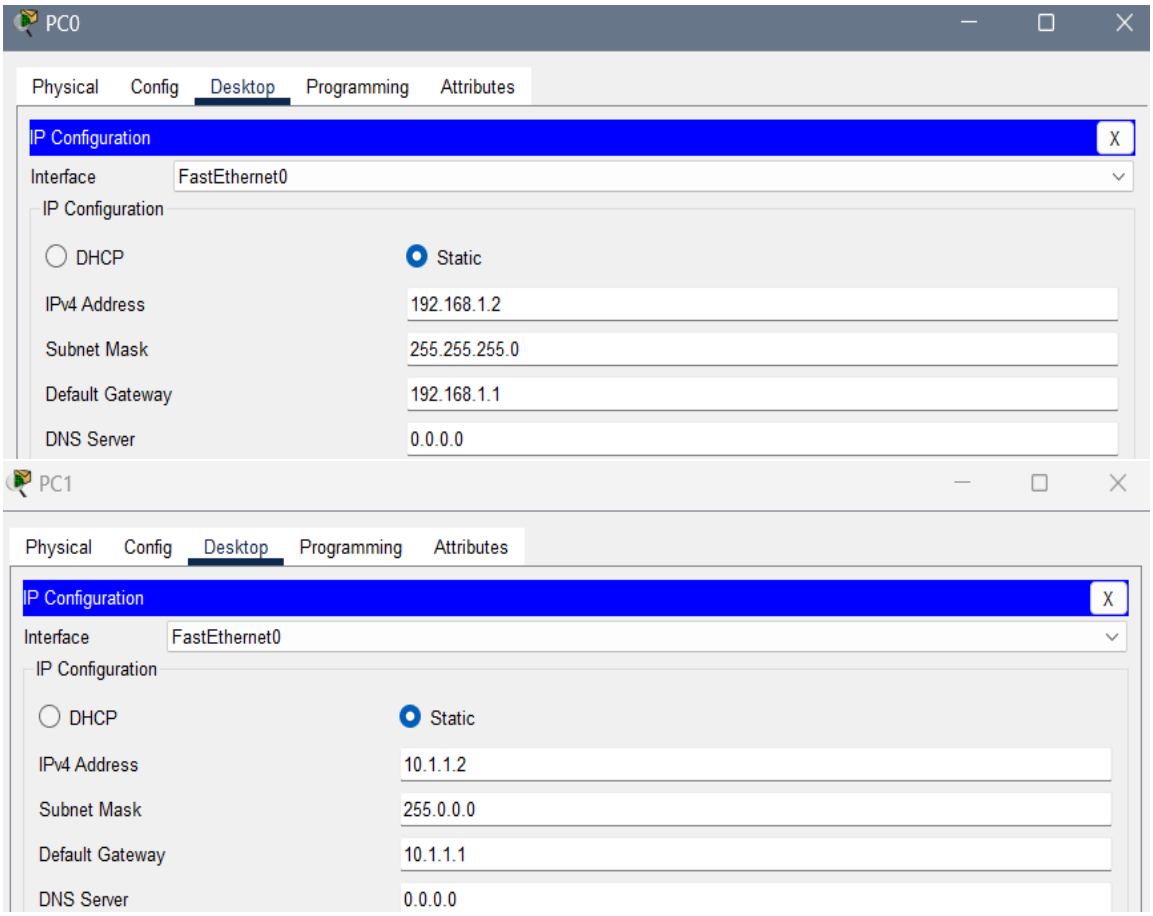
- A **router** is a device that connects two or more packet-switched networks. It uses CLI to configure the network device.
- **CLI**- A Command Line Interface. It is a software mechanism we use to interact with your operating system using your keyboard.
- Command line skills are indispensable for working with limited computing resources or cloud services. Less resource consumption and high speed.
- Router configuration is the process of setting up the parameters, rules, and features of a router.
    - The Global Configuration Mode is like a setting where the entire routers configuration is adjusted like assigning names and setting passwords.
    - Router interface configuration involves setting up and managing network interfaces on a router.

4. Procedure

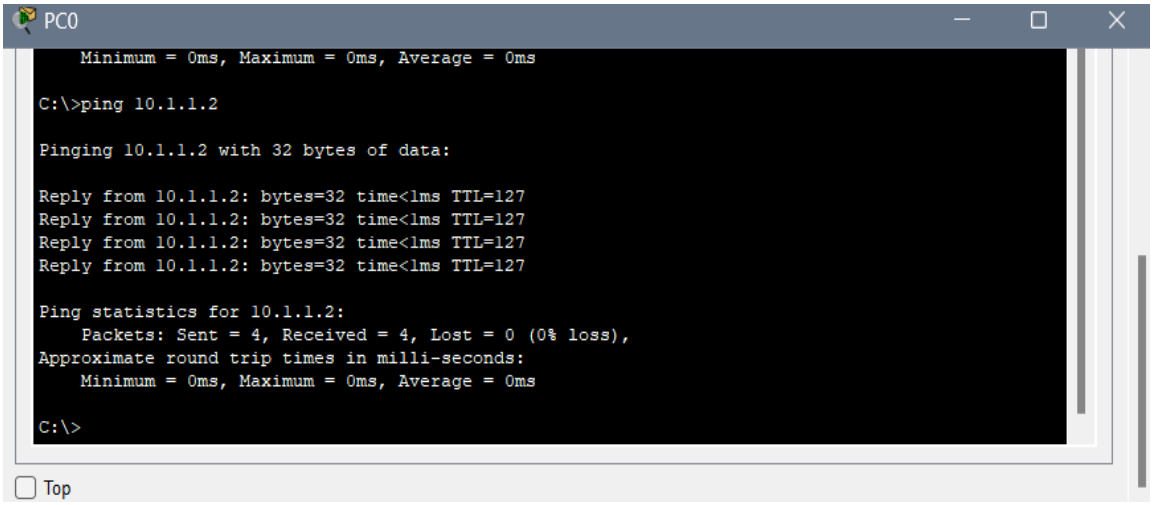
1.Router interface config with valid IP addresses



2.Host address config for **PCs** with valid IP address



3.Test that the router config is functional using **Ping** commands from **PC0** to **PC1**.



5. Conclusion

The network was successfully implemented and configured using Cisco Packet Tracer. By assigning valid IP addresses to the router's interfaces and the PCs, connectivity was established between the two hosts. Then did a ping test which verified the proper configuration of the network. Doing this practical work helps us understand about the networking concepts, including IP addressing and router configuration through CLI.

6. References

- YouTube video:

  Cisco CLI for beginners:
  https://www.youtube.com/watch?v=dtn-2QCXm-Q

  Configuring a Router using CLI in Cisco Packet Tracer:
  https://www.youtube.com/watch?v=n315uaFPcX4

- Document:

  https://www.cisco.com/c/en/us/td/docs/routers/access/800M/software/800MSCG/routconf.html